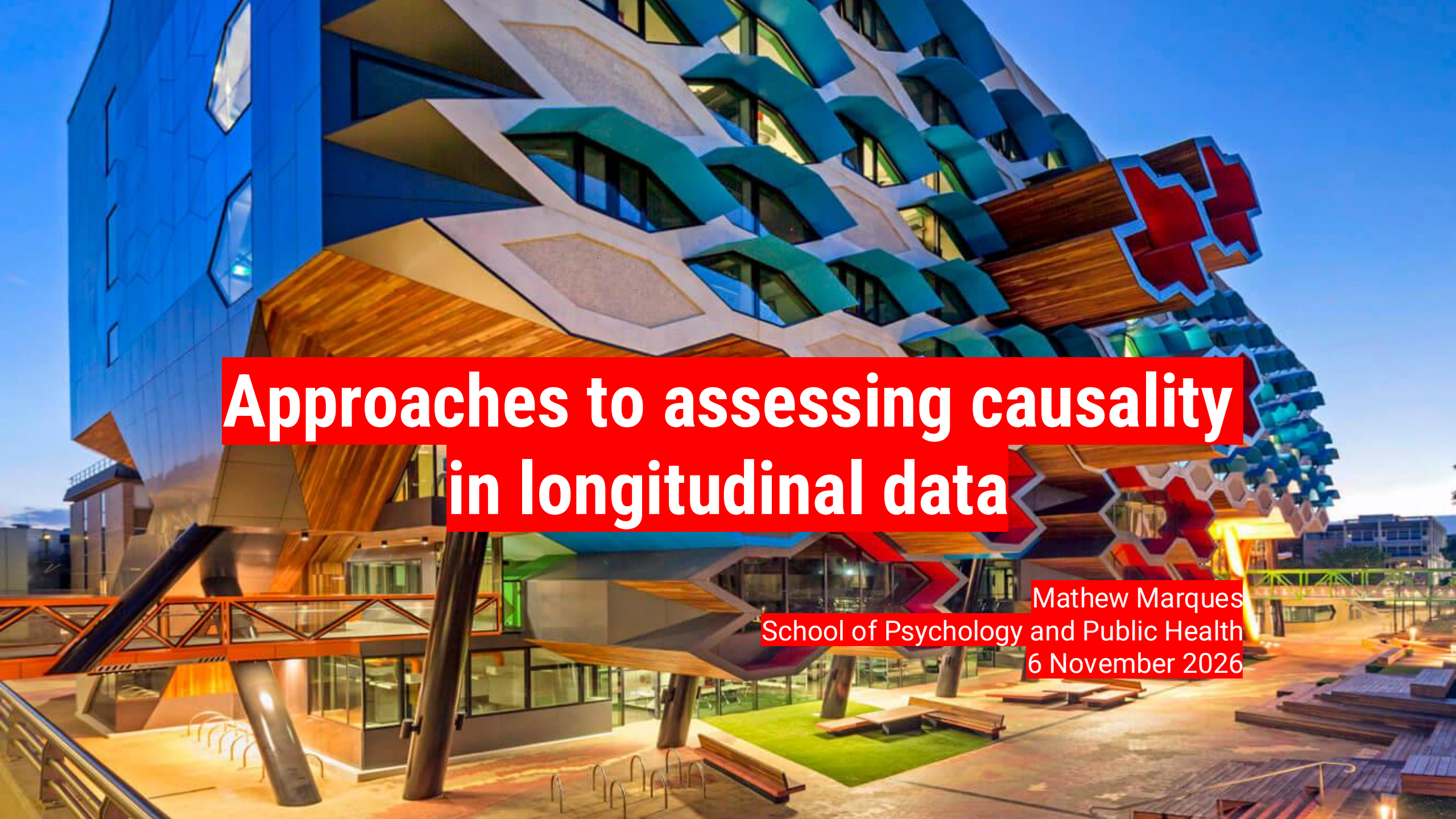


# Approaches to assessing causality in longitudinal data

Mathew Marques
School of Psychology and Public Health
6 November 2026

# Methodologies for assessing causality

Experimental design

Longitudinal design ?

Cross-lagged panel model (CLPM) (Campbell, 1963; Kenny, 1975)

Random Intercepts cross-lagged panel model (Hamaker et al., 2015)

Extensions of the RI-CLPM (Mulder & Hamaker, 2021)

# Some Logical Assumptions (Mill, 1843)

**Time precedence of the cause:** The cause has to precede the effect

**Covariation:** The cause has to be demonstrably related to the effect

There are **no plausible alternative explanations** of the effect



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# Methodological Assumptions

To compare constructs across time, need to establish measurement invariance

Configural invariance (pattern of fixed & free parameters is equivalent across time)

The same questions represent each construct at time points

Weak / metric invariance (respective factor loadings are equal across time)

Each question relates to its construct with the same strength at timepoints

Strong / scalar invariance (respective indicator means are equal across time) *

The item intercepts are equal across time: people with the same underlying level of the construct are expected to give the same average response to each at timepoints

Strict / residual factorial invariance (indicator residuals are exactly equal across time) #

Each question has the same amount of item-specific measurement error at both times

(Osborne & Little , 2024)



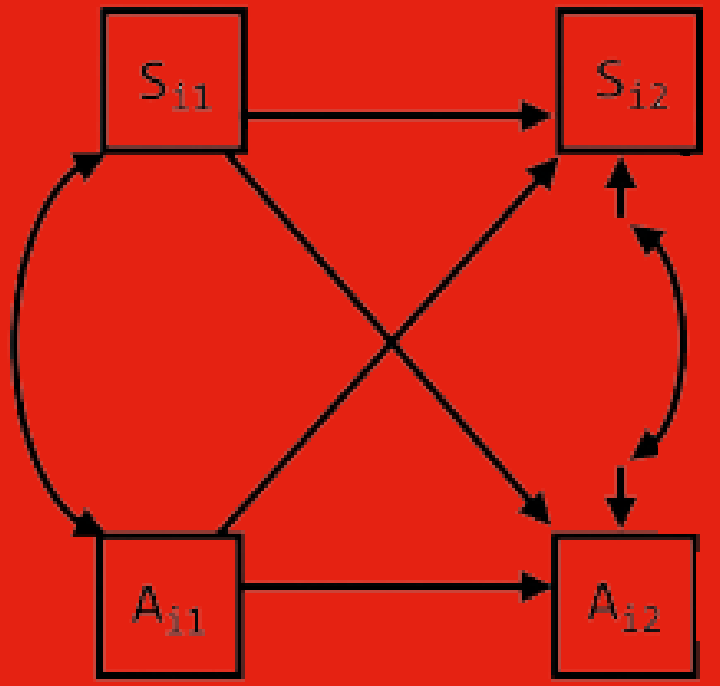
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# Cross-lagged panel model

Method to analyse panel data where two or more variables are repeatedly measured at two or more different timepoints

Four variables, there are 6 possible relations

- 2 cross-sectional correlations
- 2 auto-regressive relationships
- 2 cross-lagged relationships



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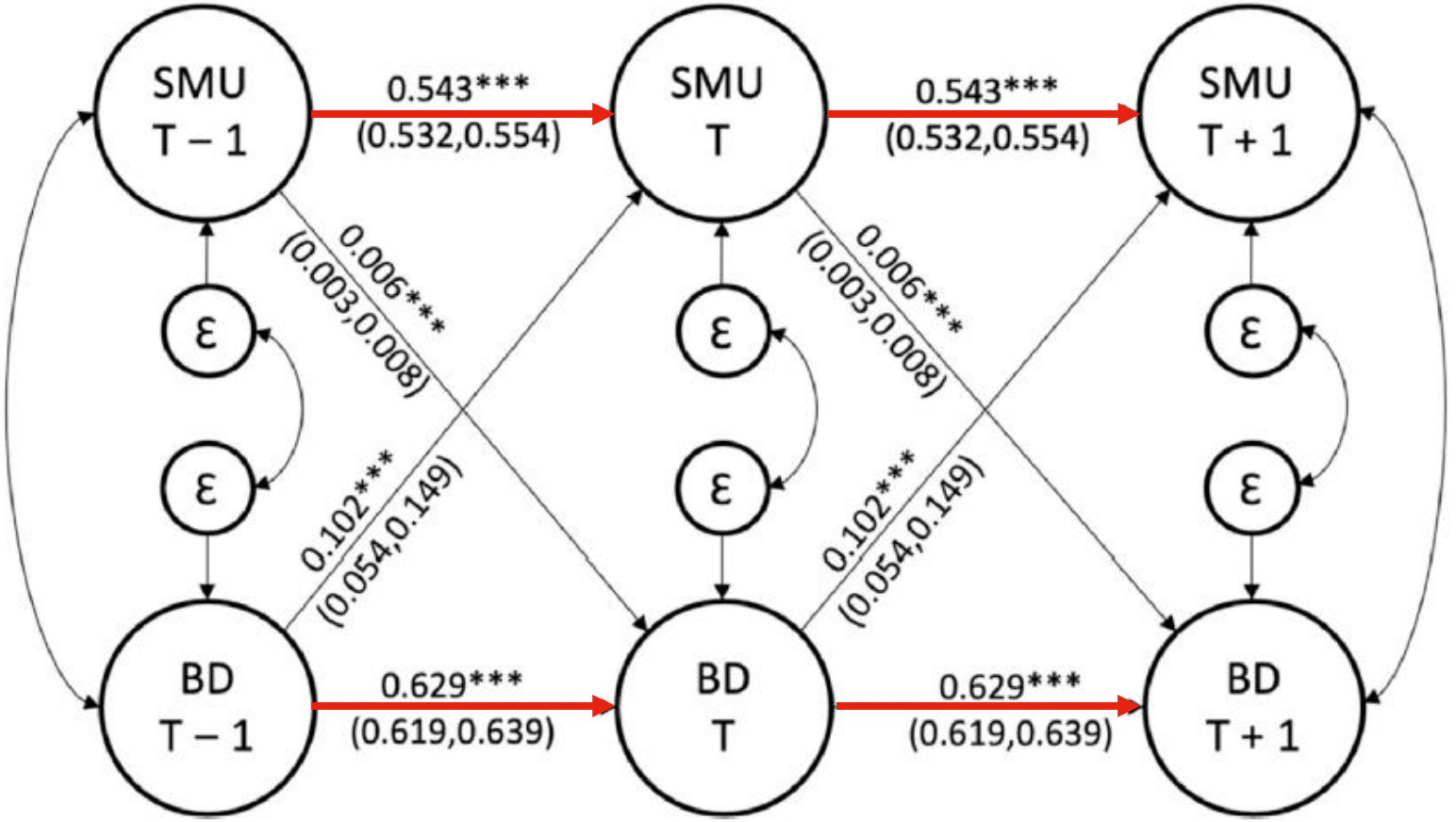
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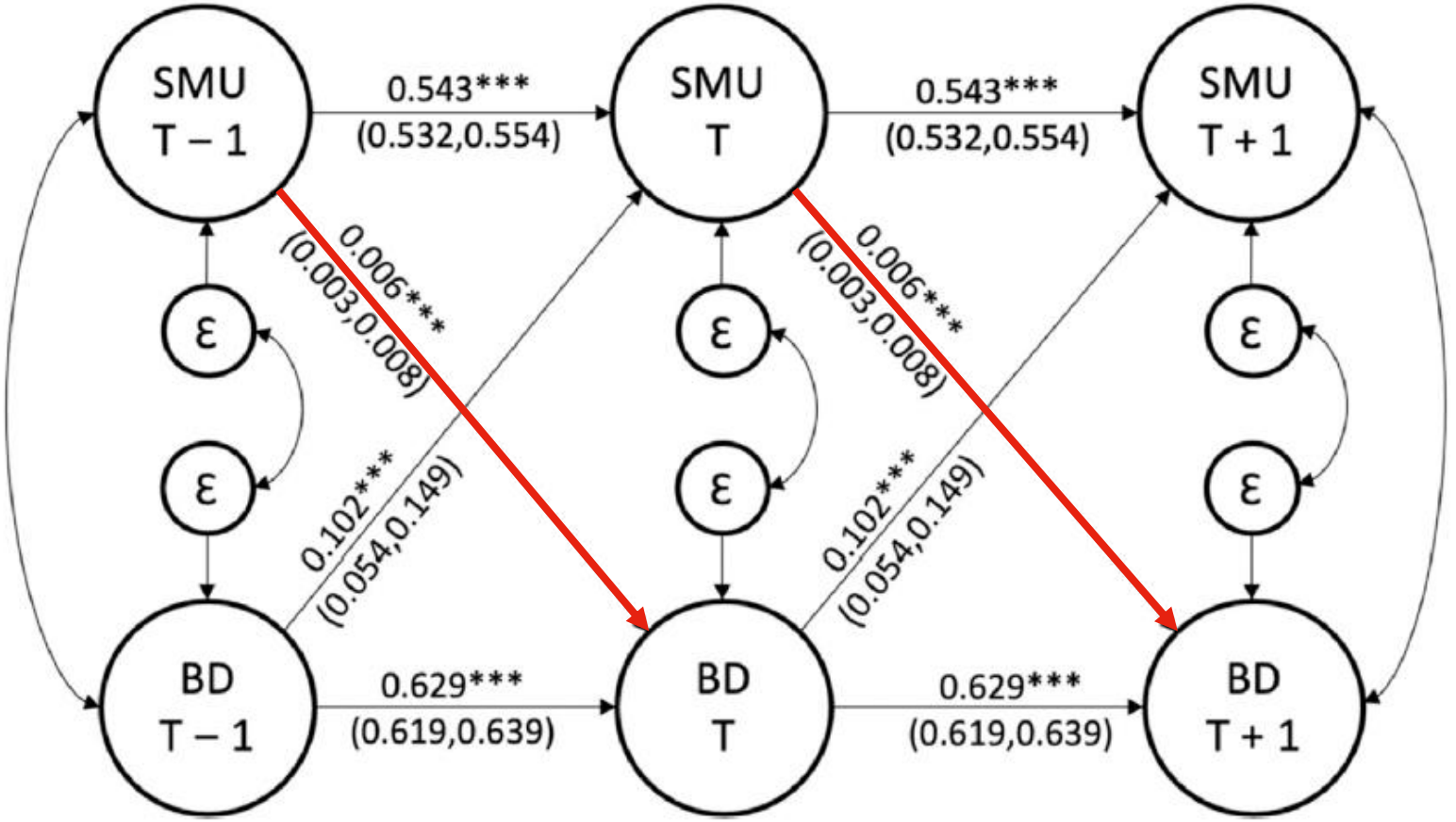
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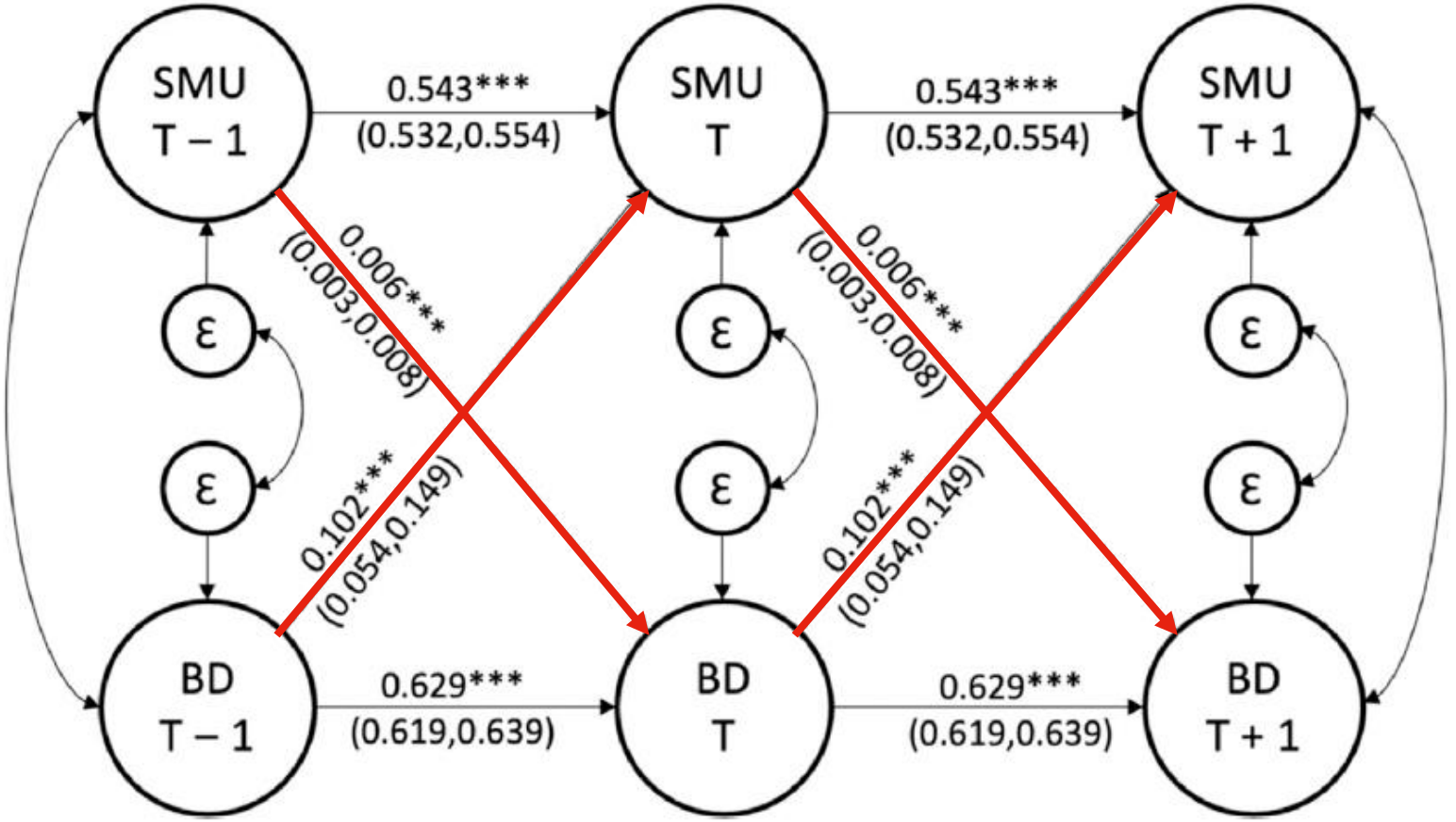
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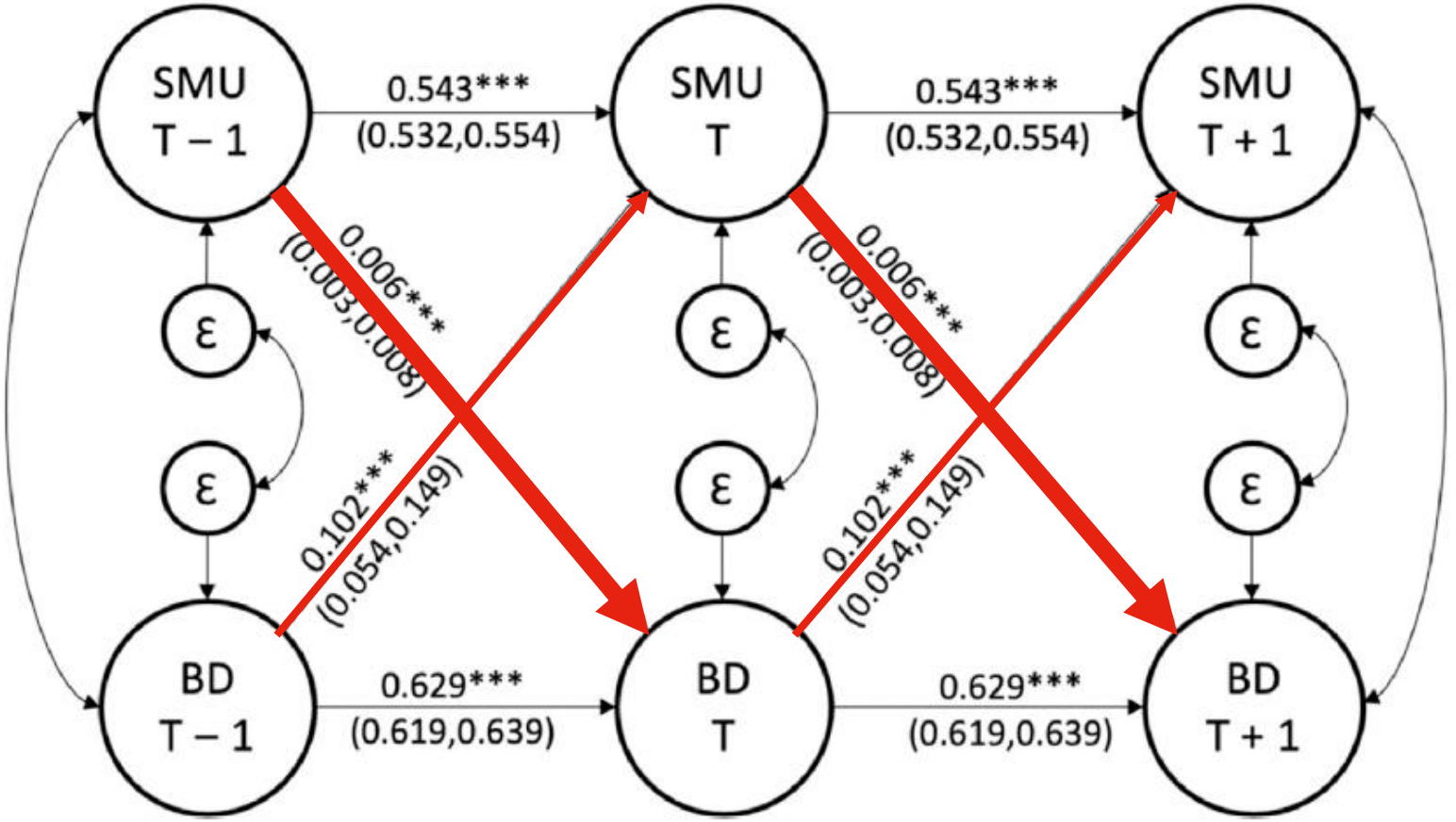
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# Cons of the CLPM

Researchers often test models that are just identified

Yields biased cross-lagged effects when constructs have a trait-like component

Confounds between-person stability and within-person change

(Osborne & Little , 2024)



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“if stability of the constructs is to some extent of a trait-like, time-invariant nature, the inclusion of autoregressive parameters will fail to adequately control for this. As a result, the estimates of the cross-lagged regression coefficients will be biased, which may lead to erroneous conclusions regarding the underlying causal pattern”

“we demonstrate the spurious results that may arise when using the cross-lagged panel model to analyze data that include stable, trait-like individual differences.”

(Hamaker et al., 2015)

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# A Critique of the Cross-Lagged Panel Model

Ellen L. Hamaker and Rebecca M. Kuiper
Utrecht University

Raoul P. P. P. Grasman
University of Amsterdam

The cross-lagged panel model (CLPM) is believed by many to overcome the problems associated with the use of cross-lagged correlations as a way to study causal influences in longitudinal panel data. The current article, however, shows that if stability of constructs is to some extent of a trait-like, time-invariant nature, the autoregressive relationships of the CLPM fail to adequately account for this. As a result, the lagged parameters that are obtained with the CLPM do not represent the actual within-person relationships over time, and this may lead to erroneous conclusions regarding the *presence*, *predominance*, and *sign* of causal influences. In this article we present an alternative model that separates the within-person process from stable between-person differences through the inclusion of random intercepts, and we discuss how this model is related to existing structural equation models that include cross-lagged relationships. We derive the analytical relationship between the cross-lagged parameters from the CLPM and the alternative model, and use simulations to demonstrate the spurious results that may arise when using the CLPM to analyze data that include stable, trait-like individual differences. We also present a modeling strategy to avoid this pitfall and illustrate this using an empirical data set. The implications for both existing and future cross-lagged panel research are discussed.

*Keywords:* cross-lagged panel, reciprocal effects, longitudinal model, trait–state models, within-person dynamics

In 1980, Rogosa's seminal article *A Critique of the Cross-Lagged Correlation* was published, which successfully conveyed the message that comparing cross-lagged correlations from longitudinal panel data is an inappropriate basis for making causal inferences.[1] One of the key insights stemming from Rogosa's article is that, if two constructs are characterized by different degrees of stability, the comparison of cross-lagged correlations may lead to spurious conclusions regarding the causal mechanism. Since then, most researchers interested in causality in panel data have abandoned cross-lagged correlations and endorsed the *cross-lagged panel model* (CLPM)—also known as the *cross-lagged path model* or the *cross-lagged regression model*—instead. In the CLPM, stability of the constructs is controlled for through the inclusion of autoregressive relationships, and it is therefore often believed that the cross-lagged regression parameters obtained with this model are the most appropriate measures for studying causality in longitudinal correlational data (e.g., Deary, Allerhand, & Der, 2009; Soenens, Luyckx, Vansteekiste, Duriez, & Goossens, 2008; Wood, Maltby, Gillett, Linley, & Joseph, 2008). Specifically, it is common practice to standardize the cross-lagged regression coefficients and compare their relative strength with determine which variable has a stronger causal influence on the other (Bentler & Speckart, 1981).

The current article forms a sequel to the warning given by Rogosa (1980), in that it will be argued that not only should we *account* for stability, but we also need to *account for the right kind* of stability. It will be shown that if stability of the constructs is to some extent of a trait-like, time-invariant nature, the inclusion of autoregressive parameters will fail to adequately control for this. As a result the estimates of the cross-lagged regression coefficients will be biased, which may lead to erroneous conclusions regarding the underlying causal pattern. This message is not novel in itself: In fact, it has been recognized repeatedly that the “omitted variable problem” may affect the estimation of the cross-lagged coefficients (e.g., Allison, 2009; Dwyer, 1983; Finkel, 1995; Heise, 1970), and diverse modeling strategies have been proposed to

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Correspondence concerning this article should be addressed to Ellen L. Hamaker, Methodology and Statistics, Faculty of Social and Behavioural Sciences, Utrecht University, P.O. Box 80140, 3508 TC, Utrecht, The Netherlands. E-mail: e.l.hamaker@uu.nl

[1] While the omitted variable problem implies that we cannot make strong causal statements based on correlational data, it does not prohibit the use of the concept of *Granger causality* (Granger, 1969). However, many researchers using cross-lagged regression refrain from using the term *causal*, and use terms like *reciprocal relationship* (Erickson, Wolfe, King, King, & Sharkansky, 2001; Lindwall, Larsman, & Hagger, 2011), *role* (Ribeiroet al., 2011), *cross-domain effects* (Burt, Obradović, Long, & Masten, 2008), *exposure* (Cole et al., 2006), *impact* (Gault-Sherman, 2012), or *influence* (Green, Furrer, & McAllister, 2011), instead. It may be argued however, that these alternative terms also imply a causal mechanism, and even more so, that an interest in causality is actually the driving force behind these studies. Therefore, we decided to use the terms *causal* and *causality* in the current article, although we acknowledging that strong causal statements can only be based on experimental designs, and we should confine ourselves to the concept of Granger causality.

# (Over) simplification of the issue

Which scenario below shows change?

… but *r* = 1.0 for both !

## Scenario 1

| ID | Time 1 | Time 2 | Time 3 |
|---|---|---|---|
| 1 | 5 | 5 | 5 |
| 2 | 3 | 3 | 3 |
| 3 | 2 | 2 | 2 |
| … | … | … | … |
| 100 | 7 | 7 | 7 |

## Scenario 2

| ID | Time 1 | Time 2 | Time 3 |
|---|---|---|---|
| 1 | 4 | 5 | 6 |
| 2 | 3 | 4 | 5 |
| 3 | 1 | 2 | 3 |
| … | … | … | … |
| 100 | 5 | 6 | 7 |

(Osborne & Little , 2024)



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# Simplification of the solution

Random intercepts account for the trait-like stability of the constructs

## Scenario 1

| ID | Time 1 | Time 2 | Time 3 | *M* |
|---|---|---|---|---|
| 1 | 5 | 5 | 5 | 5 |
| 2 | 3 | 3 | 3 | 3 |
| 3 | 2 | 2 | 2 | 2 |
| … | … | … | … | … |
| 100 | 7 | 7 | 7 | 7 |

## Scenario 2

| ID | Time 1 | Time 2 | Time 3 | *M* |
|---|---|---|---|---|
| 1 | 4 | 5 | 6 | 5 |
| 2 | 3 | 4 | 5 | 4 |
| 3 | 1 | 2 | 3 | 2 |
| … | … | … | … | … |
| 100 | 5 | 6 | 7 | 6 |

(Osborne & Little , 2024)

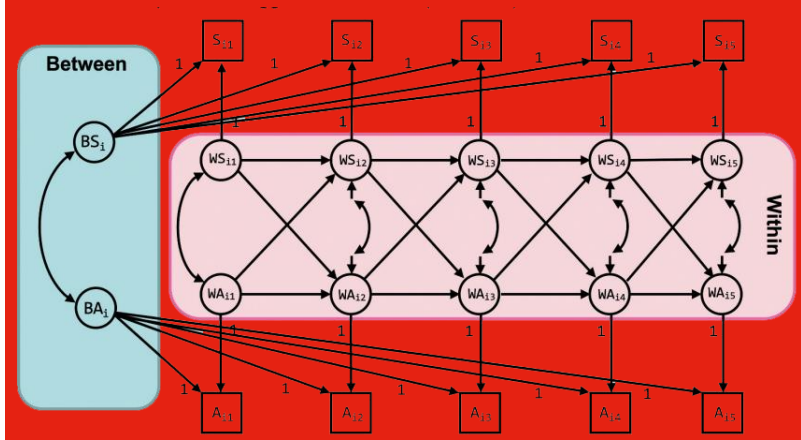


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# Random-Intercepts CLPM

Separates between-person stability (i.e., stable trait-like differences) from within-person (i.e., time variant state like) changes thus overcoming drawbacks of previous modelling techniques

- Between person components indicate on average associations over time
- Auto-regressive terms now indicate changes from a person's trait-level mean in each respective measure at the next timepoint (inertia or carry over effects)
- Cross-lagged terms indicate change from individuals' prior state – difference between observed and expected (grand mean + random intercept) score



# Sleep and psychological distress



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## Increased sleep predicts annual decreases in psychological distress: Results from a 6-year longitudinal panel sample

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ARTICLE INFO

*Keywords:*
Sleep duration
Psychological distress
Cross-lagged panel model
New Zealand

ABSTRACT

*Objective:* To investigate the annual temporal ordering of sleep quantity and psychological distress, separating between-person stability from within-person change.
*Design:* Random-intercepts cross-lagged panel model using 6 annual waves of longitudinal data from the New Zealand Attitudes Values Study postal questionnaire.
*Participants:* New Zealand Attitudes Values Study respondents in 2013, 2014, 2015, 2016, 2017, and 2018 (*N*s = 17,880; 15,757; 13,904; 21,849; 17,031; and 47,462).
*Measurements:* Participants were asked, "During the past month, on average, how many hours of actual sleep did you get per night?" and responded to the K6 psychological distress scale each year. They also reported their demographic characteristics.
*Results:* Identified longitudinal associations between sleep duration and psychological distress in a traditional cross-lagged panel model were mostly attributable to the stability of the between-person differences in sleep duration and psychological distress. We provide evidence to suggest that increased sleep duration as indicated over a short period of time (ie, 1 month) predicted lower within-person levels of psychological distress the following year. Psychological distress did not predict sleep duration, in contrast.
*Conclusions:* Our analyses suggest that sleep duration in this sample of New Zealanders precedes psychological distress. This is significant given the propensity for short sleep in this sample and issues of poor mental health and short sleep among low SES indigenous members of this community. The promotion of adequate sleep duration may yield positive gains in psychological well-being.
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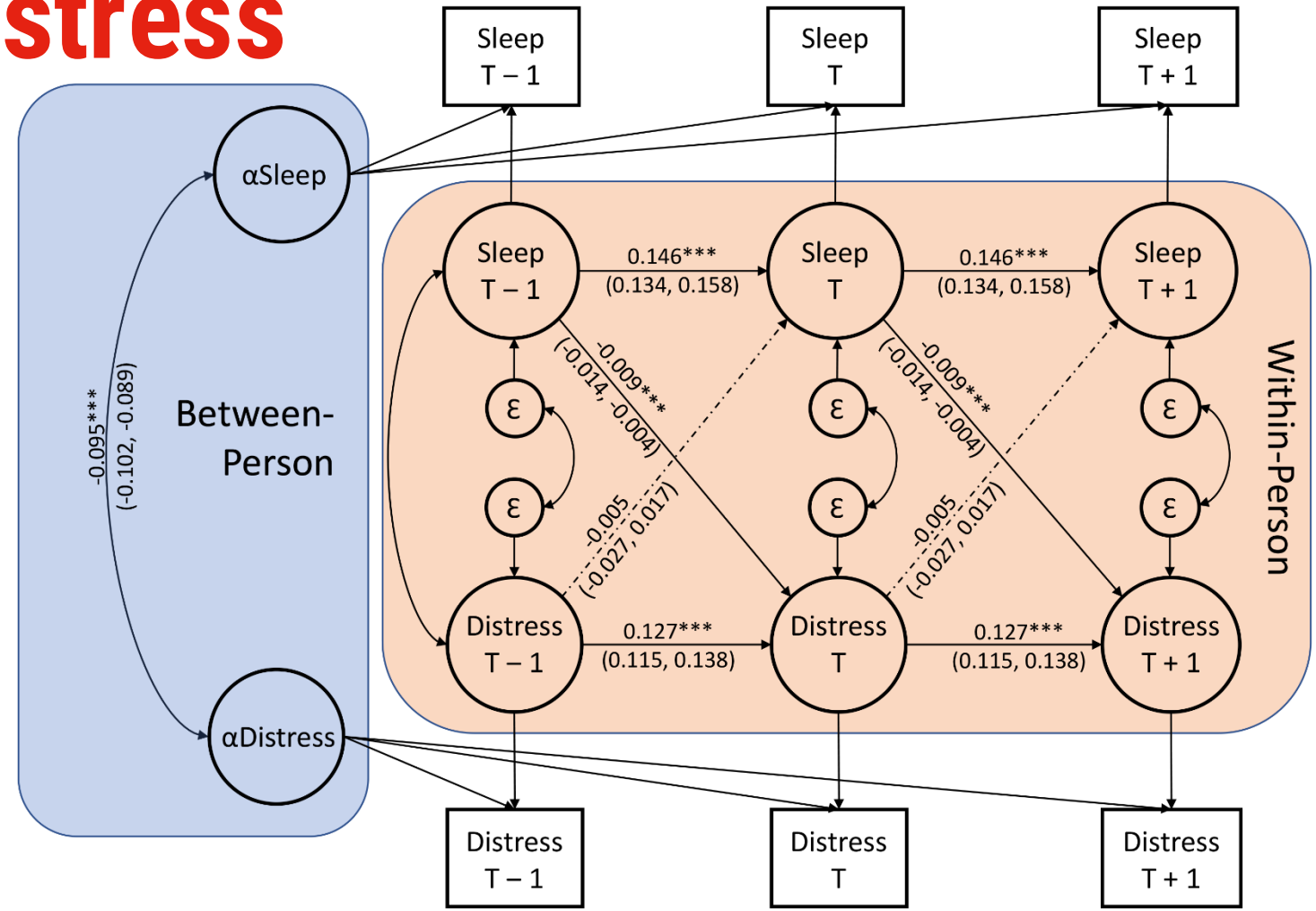
**Introduction**

Sleep disturbances such as insomnia, sleep apnea, and narcolepsy are related with the incidence of mood disorder diagnoses (eg, major depression, bipolar disorders, and cyclothymic disorder).[1] Emerging evidence from large community prospective investigations reveal relationships between short sleep duration with self-reported psychological distress[2] and psychological distress with short sleep duration.[3] Meta analyses[4] from other prospective studies confirmed associations of both insufficient[5–10] and long sleep[7–10] duration with negative affect (ie, frequency of depressive and anxious emotions). While these large data sets have been useful, differences in study design, measurement of sleep and negative affect, quantification of "long" and "insufficient" sleep, and statistical approach, have hindered conclusions about the direction of association between sleep duration and negative affect. Given the costs to both the individual and society associated with these conditions, an understanding of the temporal order of the sleep duration-negative affect pathway is required to identify individuals "at risk" and to inform public health campaigns and evidence-based intervention.

A systematic review[1] of 9 studies (8 prospective, *N* = 7336) reported a bidirectional relationship between sleep disturbance and mood disorders in 4 studies, a unidirectional relationship between sleep disturbance and depression in 3 studies and no relationship between both variables in one study. The authors concluded there is a bidirectional relationship between insomnia and depression. While there is evidence of an association between sleep disorders and mood disorders, there is a dearth of research that has assessed the directional pathway between sleep duration and nonclinical levels of negative affect in community samples. A measure of nonspecific psychological distress such as the Kessler psychological distress scale (K6) is frequently used in community-based epidemiological research as it is a measure that captures elements of primarily depressive and anxiety-

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# Interpretations between CLPM and RI-CLPM

| | TRADITIONAL CLPM | RI-CLPM |
|---|---|---|
| Autoregressive effect ($X_{T-1} \rightarrow X_T$) | Rank order stability of a construct (after adjusting for the cross-lagged effect) over time.<br><br>"Those who are relatively high on sleep minutes at one time point tend to be relatively high on sleep minutes at a later timepoint". | Tendency for deviations from a trait-level mean to carry over into a later timepoint (after adjusting for the cross-lagged effect); often referred to as inertia.<br><br>"A deviation from a person's trait level sleep duration persists as a deviation from their trait level sleep duration at a later timepoint". |
| Cross-lagged effect ($X_{T-1} \rightarrow Y_T$) | Extent to which one's relative position on variable X at T-1 predicts change in one's relative position on Variable Y at a later timepoint (after adjusting for the stability of Variable Y).<br>"Relative to those lower on sleep minutes, those who are higher on sleep minutes show a larger increase in distress at a later timepoint". | Extent to which a deviation from a person's average score on Variable X predicts a deviation from their average score on Variable Y at a later assessment (after adjusting for inertia in Y).<br><br>"When a person's sleep duration decreases below their trait level, they display an increase above their trait level in distress at a later point in time". |

(Osborne & Little , 2024)



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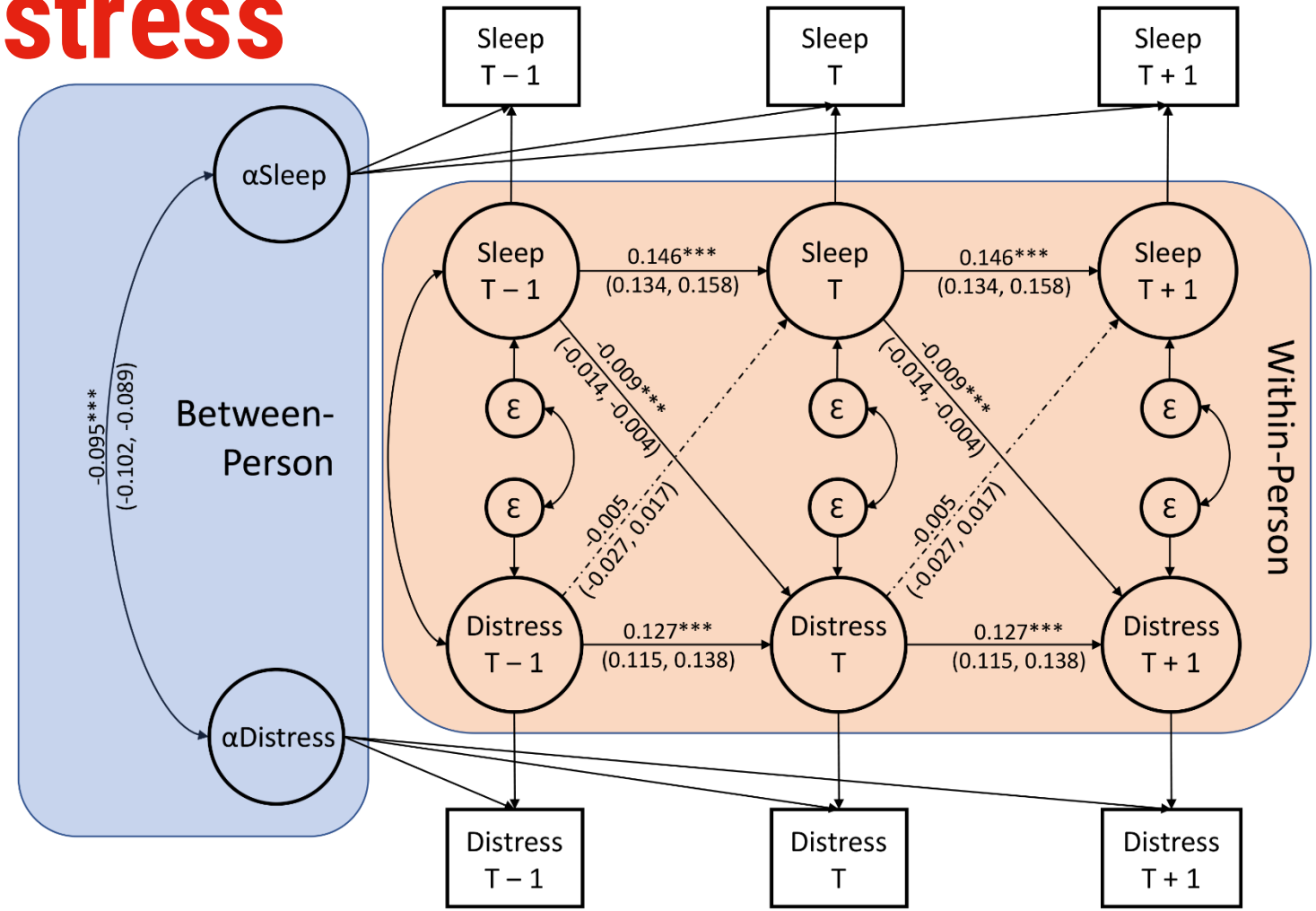
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# Thoughts and considerations



CLPM requires 2T; RI-CLPM >2T

Intervals between T are equal

Can constrain lagged parameters over time

Possible to consider >2 vars @ each T

Multigroup models are possible

How to handle missing data – FIML (Enders, 2001; Enders & Bandalos, 2001)

# RI-CLPM Pros and Cons

Requires relatively few waves of data (three or more)
Separates variance into between-person and within-person components
Conceptually similar to CLPM
Fits data better than traditional CLPMs (CLPM is nested under RI-CLPM)

Autoregressive coefficients are counter-intuitive (inertia, not stability)
Effects tend to be small
Ignores between-person change

(Osborne & Little , 2024)



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# Resources

RI-CLPM and extensions in *R* and *Mplus*
https://jeroendmulder.github.io/RI-CLPM/index.html

Osborne, D., & Little, T. D. (2024). The random intercept cross-lagged panel model. In *Longitudinal structural equation modeling* (Second edition, pp. 286–337). The Guilford Press.

# Knowledge → action

Let's use *R* and *lavaan* to recreate the RI-CLPM (and CLPM) presented earlier on sleep minutes and distress.



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# CLPM vs. RI-CLPM
# When does it matter?

## High degree of between-person stability

**Table 3**
Path coefficients of the associations between sleep duration and psychological distress

| | | Traditional cross-lagged | | Random intercepts cross-lagged | |
|---|---|---|---|---|---|
| | | Panel model | | Panel model | |
| Outcome | Predictor $_{T-1}$ | *B (SE)* | 95% CI | *B (SE)* | 95% CI |
| Sleep duration | | | | | |
| | Sleep duration | 0.631 (0.002)* | (0.627, 0.636) | 0.146 (0.006)* | (0.134, 0.158) |
| | Psychological distress | −0.062 (0.004)* | (−0.070, −0.053) | −0.005 (0.011) | (−0.027, 0.017) |
| Psychological distress | | | | | |
| | Psychological distress | 0.751 (0.002)* | (0.747, 0.755) | 0.127 (0.006)* | (0.115, 0.138) |
| | Sleep duration | −0.012 (0.001)* | (−0.014, −0.009) | −0.009 (0.003)* | (−0.014, −0.004) |
| Slope contrast | | 0.050 (0.005)* | (0.040, 0.059) | 0.004 (0.011) | (−0.017, 0.025) |
| Model fit | | | | | |
| $\chi^2$ | | | 16457.234 | | 526.573 |
| *df* | | | 56 | | 53 |
| *P* | | | <.0001 | | <.0001 |
| CFI | | | 0.841 | | 0.995 |
| SRMR | | | 0.132 | | 0.022 |
| RMSEA | | | 0.071 (0.070, 0.072) | | 0.012 (0.011, 0.013) |

Notes. Slope contrast refers to difference between cross-lagged effects in the given model.
*$P < .001$.



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# Thanks to

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Dr Siân McLean

Dr Hannah Jarman

Dr Brad Wright

Dr Carol Lee

Professor Chris Sibley

Associate Professor Danny Osborne






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# Thank You

# Questions?

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